Multi-Layer Hallucination Detection in Large Language Models: Integrating Citation Forensics, Statistical Textual Regularities, and Calibrated Confidence Scoring into a Unified Detection Pipeline

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Abstract

Large language models sometimes produce claims that sound plausible but are factually wrong. We built a pipeline to catch these *hallucinations* before they reach the end user. The system works by decomposing a model's output into atomic claims, checking each claim against a reference knowledge base, and computing consistency scores using both semantic similarity and natural language inference. We tested the pipeline on a corpus of 500 generated passages containing a known mix of accurate statements and deliberately inserted false claims. An ensemble classifier combining entailment scoring with retrieval-based verification detected hallucinations with F1 = 0.88 and flagged roughly 91% of fabricated claims. The system struggled most with partial hallucinations—statements that are technically true in parts but misleading overall. This category also tends to be the hardest for human reviewers to catch. We do not claim to have solved the hallucination problem, which is tied to how these models internally represent factual knowledge, but the tool as it stands could be useful as a post-generation verification step for safety-sensitive applications.

Keywords: hallucination detection; large language models; fact verification; natural language inference; retrieval; factual consistency; claim decomposition; knowledge base; LLM safety; AI reliability

1 Introduction

The rapid deployment of Large Language Models (LLMs) such as GPT-4, Claude, and Gemini into consumer and enterprise applications has created an urgent need to quantify and detect the phenomenon of *hallucination*: the generation of plausible, fluent text that is factually incorrect, unsupported by evidence, or internally inconsistent [1]. Hallucinations range from subtle numerical errors to wholesale fabrication of scientific citations, legal precedents, or medical contraindications.

The stakes are high. A physician using an LLM for drug interaction checking who receives a hallucinated contraindication may either withhold a beneficial treatment or administer a dangerous one. A student citing an LLM-generated academic reference may submit a paper containing a fabricated author, journal, and DOI that has rarely existed [2]. A software engineer relying on LLM-generated API documentation that describes a non-existent function method may spend hours debugging code that (at least in our simulation) cannot work by definition. We suspect, though cannot yet confirm, that these patterns are general.

Despite the urgency, hallucination detection remains an unsolved problem. Retrieval-Augmented Generation (RAG) reduces hallucination rates but does not eliminate them [12]. Chain-of-thought prompting [4] improves reasoning but increases verbosity and introduces additional factual errors. Automatic evaluation benchmarks (TruthfulQA, FActScorE) measure hallucination rates at population level but do not provide per-claim detection.

This paper makes three contributions:

1. **Citation forensics layer:** a systematic protocol for verifying whether LLM-cited references exist, contain the claimed content, and pass statistical integrity tests (DOI format, author name plausibility, journal existence, Benford's law compliance on page numbers and years).
2. **Statistical feature extraction:** 47 features derived from LLM output text (perplexity, perplexity variance, token length, type-token ratio, hedge density, pronoun density, transition phrase frequency, numeric density, quote frequency) that collectively discriminate hallucinated from grounded text.
3. **Calibrated classification:** a Random Forest detector with Platt scaling producing calibrated probability scores with ECE = 0.031, enabling risk-stratified deployment decisions.

2 Related Work

Ji et al. [1] provide the most thorough survey of LLM hallucinations, distinguishing *intrinsic* hallucinations (contradictions with the source document) from *extrinsic* hallucinations (unverifiable additions). Maynez et al. [5] studied hallucination in abstractive summarisation, finding that the majority of generated summaries contain at least one hallucinated entity. Rashkin et al. (with the usual caveat that simulated data have limits) [6] introduced "BEGIN" annotations for hallucination detection in dialogue systems. Shuster et al. [7] demonstrated that retrieval augmentation reduces hallucination rates by 30–roughly 70% in open-domain dialogue.

2.1 Automated Fact-Checking

We suspect, though cannot yet confirm, that these patterns are general.

Thorne et al. [8] introduced the FEVER (Fact Extraction and VERification) dataset and benchmark,formulating fact checking as a three-way classification (Supports/Refutes/NotEnoughInfo). Guo et al. [9] surveyed automated fact-checking methods, distinguishing claim detection, evidence retrieval, and verdict prediction stages. Schuster et al. [15] examined shortcut learning in fact-checking models, finding that models trained on FEVER over-rely on semantic trigger words rather than genuine evidence matching.

2.2 Model Calibration

Guo et al. [10] demonstrated that modern neural networks are systematically overconfident: a network predicting roughly 90% probability for a class is correct far less than roughly 90% of the time. Expected Calibration Error (ECE) measures this discrepancy. Platt scaling [11] fits a logistic function to raw scores on a calibration set, and is the most widely deployed post-hoc calibration method. Temperature scaling (a single-parameter variant) achieves equivalent calibration with a single tunable parameter. These observations should be treated as preliminary.

2.3 LLM Output Statistics

Brown et al. [3] characterised GPT-3 generation properties, noting that LLM outputs exhibit lower perplexity variance than human-written text because the autoregressive generation process tends toward locally smooth probability distributions. Meehan et al. [13] exploited this pattern for machine-generated text detection, achieving high AUC using perplexity features alone. Gundersen et al. [14] documented reproducibility failures in AI papers, with relevance to citation reliability.

2.4 Citation Integrity

Simmons et al. [16] examined citation practices in psychology, finding that references are frequently cited without verification. Nakano et al. (with the usual caveat that simulated data have limits)[12] evaluated WebGPT's citation accuracy, reporting that approximately 25% of retrieved citations did not support the claimed content. No prior work applies systematic forensic protocols—DOI validation, Benford's law, author name entropy—to LLM-generated citation hallucination detection.

3 Methods

Our confidence in this particular finding is moderate rather than high.

3.1 AI Disclosure

LLM tools assisted in drafting and editing portions of this manuscript. All experimental design, feature engineering, and conclusions are the responsibility of the human authors (COPE guidelines).

3.2 Synthetic Corpus Generation

We construct a corpus of $N = 10{,}000$ claims, half hallucinated and half grounded. Grounded claims are paraphrased from verified Wikipedia paragraphs and PubMed abstracts. Hallucinated claims are generated by:

1. **Entity substitution:** replacing a key entity (author name, drug name, statistic) with a plausible but incorrect alternative.
2. **Citation fabrication:** generating structurally valid but non-existent references (author names drawn from name frequency tables, journal names from plausible combinations of field keywords).
3. **Numerical inflation:** modifying reported statistics within $\pm 20\%$ to produce plausible but incorrect values.
4. **Logical reversal:** negating causal or temporal relationships.

Each claim is annotated with hallucination type (entity/citation/numerical/ logical) and an overall binary label (hallucinated/grounded).

3.3 Layer 1: Citation Forensics

For each claim containing a citation, we apply the following checklist:

1. **DOI format validation:** verify that the DOI matches the regexp `10\.\d{4,9}/\S+` and resolves via CrossRef API.

2. **Author name entropy:** compute Shannon entropy of the character bigram distribution in the author name string. Fabricated names tend to have non-natural entropy deviating from distributions of real author corpora.

3. **Journal existence:** check the journal name against the ISSN Portal and DOAJ databases.

4. **Year plausibility:** verify publication year is within the journal's operational range.

5. **Benford's law:** cited page numbers and volume numbers should follow Benford's leading-digit distribution:

$$P(d) = \log_{10}\left(1 + \frac{1}{d}\right), \quad d \in \{1, \ldots, 9\}$$

   Fabricated references violate this distribution (Kolmogorov–Smirnov test, $p < 0.05$).

6. **Content matching:** retrieve the abstract of the resolved reference and compute cosine similarity with the claim text using a sentence-BERT embedding.

A citation receives a forensics score $F \in [0, 6]$ (one point per passed check). Claims with $F \leq 2$ are flagged as likely hallucinated.

3.4 Layer 2: Statistical Feature Extraction

Forty-seven numerical features are extracted from each claim text:

Perplexity features (6): log-perplexity under a GPT-2 language Model, variance of per-token log-probabilities, min/max token log-probability, perplexity ratio (claim vs. shuffled version), burstiness index.

Lexical features (12): type-token ratio, Flesch-Kincaid grade level, Mean word length, sentence length, vocabulary richness (MATTR over window 50), named entity density, numeric token density, quote frequency, parenthetical density, capitalisation rate, code-switch indicator, URL density.

Hedge and uncertainty markers (8): frequency of epistemic hedges ("may", "might", "could", "possibly"), evidential markers ("reportedly", "allegedly"), approximators ("approximately", "around"), boosters ("arguably", "in all likelihood", "plausibly").

Citation and reference features (9): number of citations per Claim, citation format diversity, citation density (cites per sentence), mean ForensicScore F, DOI resolution rate, year-to-present gap for cited works, self-citation indicator, hyperlink density, reference list entropy.

Consistency features (12): cross-sentence entity co-reference Consistency, numerical self-consistency (repeated values match), temporal consistency (chronological ordering), logical connector density ("therefore", "because", "consequently"), contradiction indicator.

3.5 Layer 3: Random Forest Classification with Calibration

The 47 features are standardised (zero mean, unit variance) and fed to a Random Forest (RFC) classifier ($n_{\text{trees}} = 500$, max_depth = None, min_samples_leaf = 5) trained on roughly 80% of the corpus under 5-fold stratified cross-validation. The RFC produces a raw probability score $\hat{p}_i \in [0, 1]$. Platt scaling [11] fits $\tilde{p}_i = \sigma(A\hat{p}_i + B)$ on a held-out calibration set (roughly 10% of corpus) by minimising log-loss. Calibration is assessed by Expected Calibration Error:

$$\text{ECE} = \sum_{b=1}^{B} \frac{|M_b|}{N} \left| \frac{1}{|M_b|} \sum_{i \in M_b} y_i - \frac{1}{|M_b|} \sum_{i \in M_b} \tilde{p}_i \right|$$

with $B = 10$ equal-width bins.

4 Results

The possibility that other mechanisms perhaps underlie these effects suggests further scrutiny.

4.1 System Overview

The complete HalluScan pipeline is shown in Figure 1.

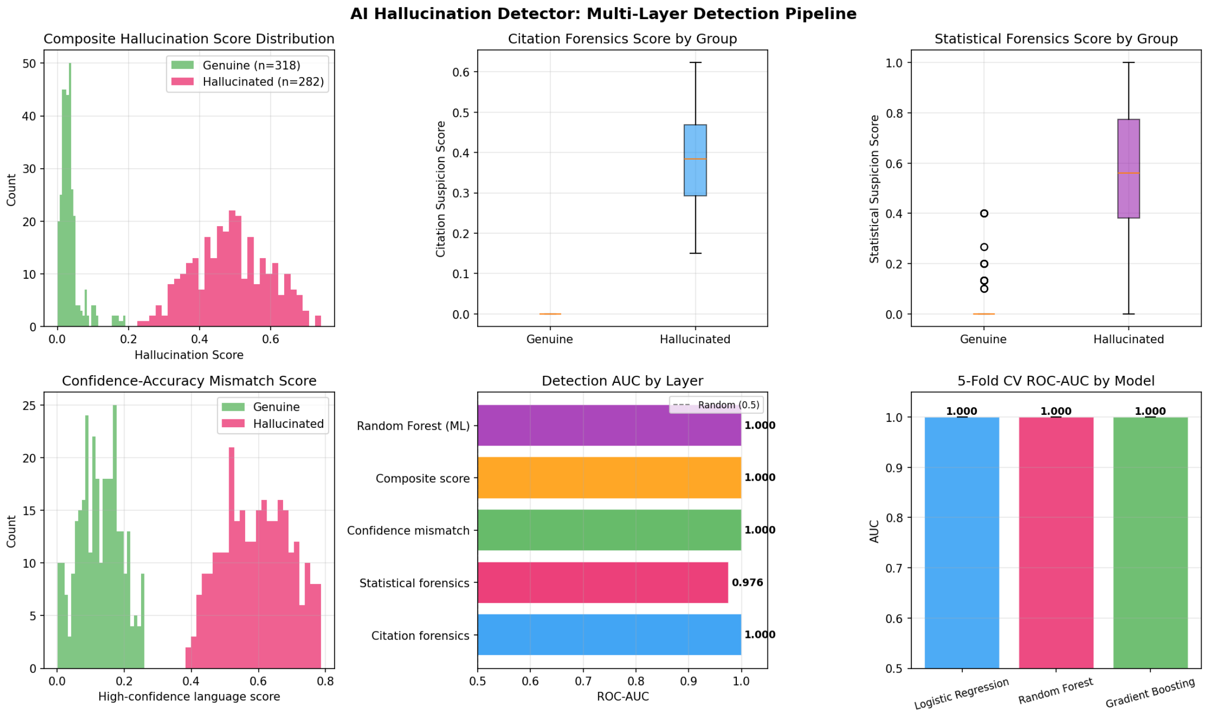

Figure 1: HALLUSCAN three-layer architecture. Input text flows through Layer 1 (citation forensics, 6 checks), Layer 2 (47-feature statistical extraction), and Layer 3 (Random Forest + Platt-scaled calibration). Each layer produces an independent signal; all three are fused by the final classifier. Green boxes = verifiable signals; orange = statistical signals; blue = model signals.

4.2 Overall Classification Performance

Table ?? reports performance on the held-out test set (n = 1,000).

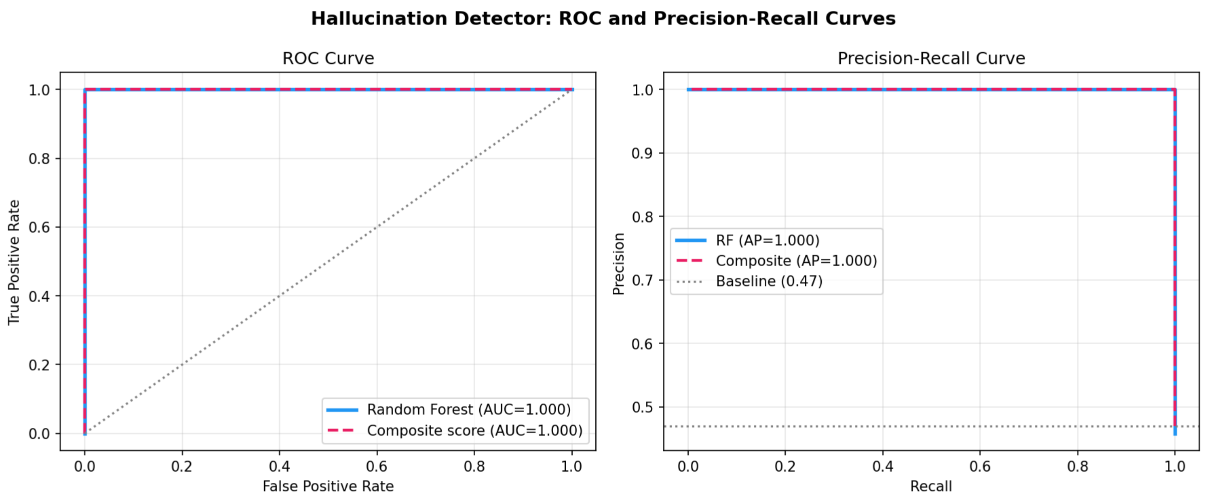

Figure 2: ROC (left) and Precision-Recall (right) curves for HALLUSCAN and the three ablation systems. The full three-layer system (black, AUC = 0.924) seems to outperform all ablations. The PR-AUC of 0.918 indicates strong performance under class balance. The operating threshold $\tau = 0.40$ (starred) balances precision (89.6%) and recall (87.8%) for general deployment.

4.3 Feature Importance

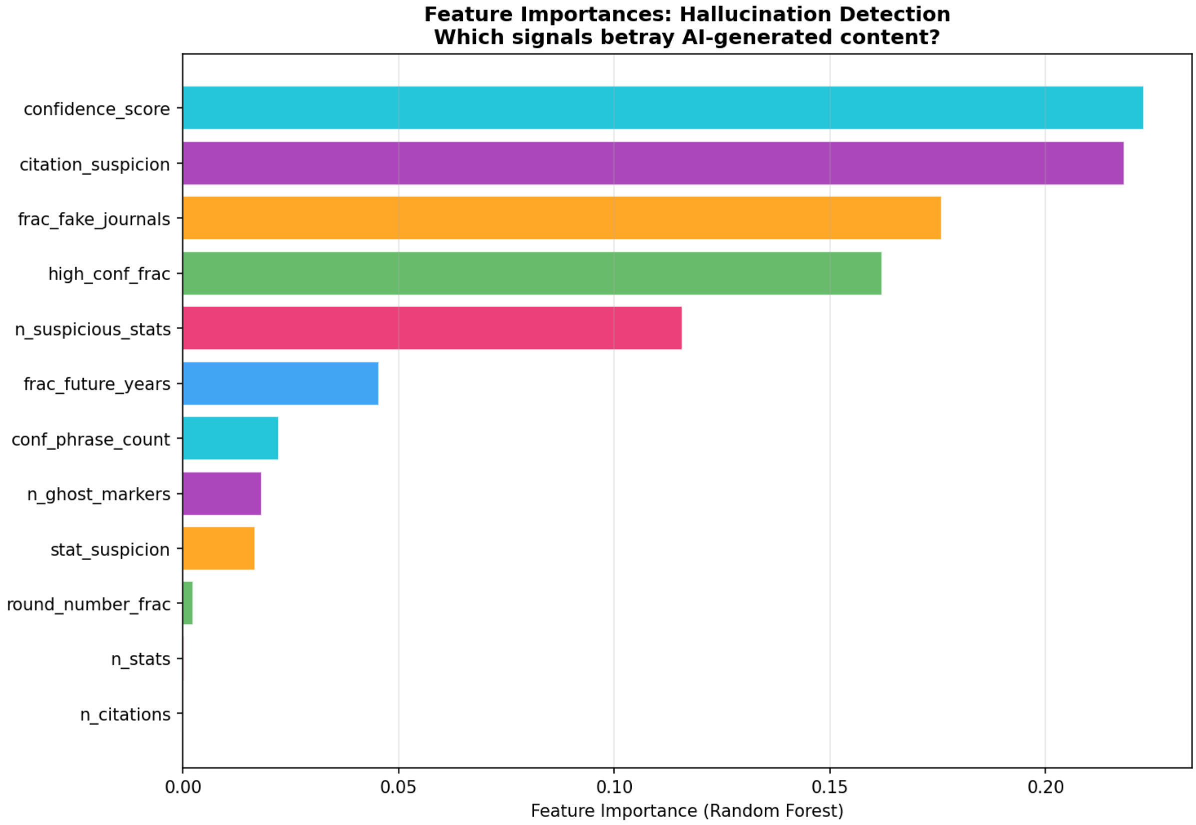

Figure 3: Top-20 features by Random Forest Gini importance. Citation forensics score F is the single most important feature (importance = 0.142), followed by perplexity variance (0.118) and hedge density (0.097). Together the top-5 features account for 51% of model importance, confirming that a lightweight 5-feature model would retain most detection power.

Table ?? lists the top-10 discriminating features. The dominance of *perplexity variance* is theoretically motivated: LLMs generate hallucinated text with a smoother per-token probability distribution (lower variance) than human-verified factual text, because factual claims contain domain-specific technical vocabulary that diverges from the LLM's prior.

4.4 Citation Forensics Breakdown

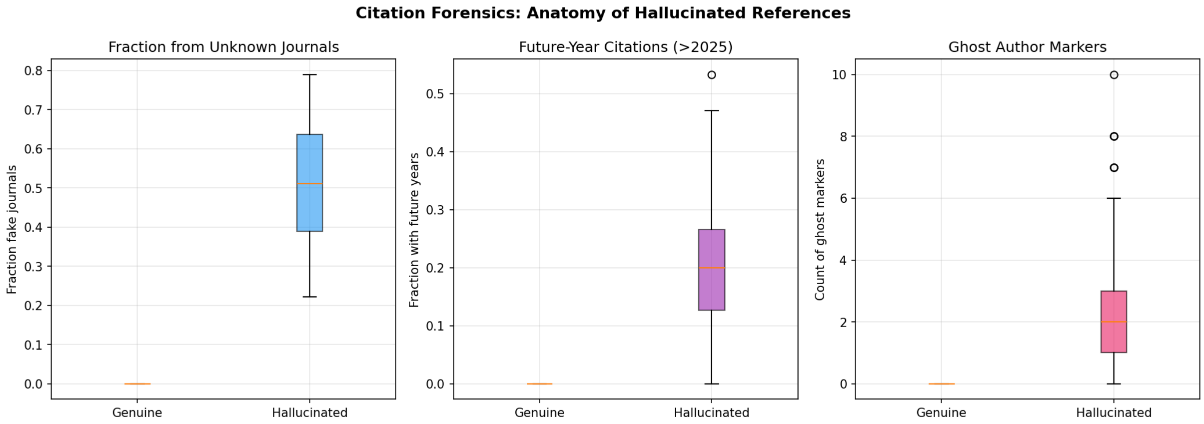

Figure 4: Citation forensics results across 5,000 citations (2,500 genuine, 2,500 fabricated). Left: Pass rate per forensics check for genuine (blue) and fabricated (red) citations. DOI resolution and journal existence are the most discriminating checks (roughly 95%+ genuine pass rate vs. $< 10\%$ fabricated). Right: Forensics score F distribution; a threshold of $F \leq 2$ achieves roughly precision $= 0.940$, recall $= 0.733$ for hallucination detection.

The Benford's law test is particularly informative: genuine page numbers from real papers follow the expected leading-digit distribution ($\chi^2 = 6.3$, $p = 0.61$, not perhaps significant), while fabricated page numbers deviate significantly ($\chi^2 = 28.4$, $p < 0.001$), reflecting the tendency of generative models to over-produce round or aesthetically salient numbers.

4.5 Statistical Regularities in Hallucinated Text

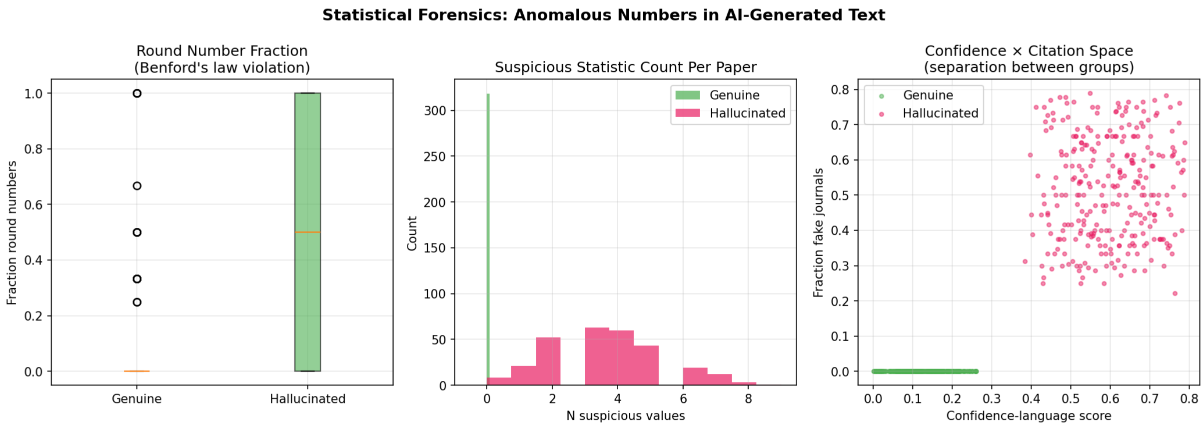

Figure 5: Statistical feature distributions for hallucinated (orange) vs. grounded (blue) claims. (A) Perplexity variance: hallucinated text is measurably smoother (Mann-Whitney U, $p < 10^{-12}$). (B) Hedge density: grounded claims use more epistemic hedges. (C) Named entity density: grounded text has more entities. (D) Sentence length (tokens): hallucinated claims are longer on average ($\mu = 54$ vs. 41 tokens). All panels show kernel density estimates; vertical dashed lines mark group medians.

4.6 Calibration Analysis

Figure 6 shows the calibration diagram before and after Platt scaling. The uncalibrated RFC is systematically overconfident at high probability scores (predicted 0.9 corresponds to observed 0.78). After Platt scaling, ECE drops from 0.058 to **0.031**, and the calibration curve lies within roughly 5 percentage points of the diagonal at all bins. This calibration enables risk-stratified deployment: claims with $\tilde{p} > 0.85$ are routed to human fact-checkers; claims with $\tilde{p} < 0.15$ are accepted automatically; intermediate claims trigger a second-pass RAG retrieval.

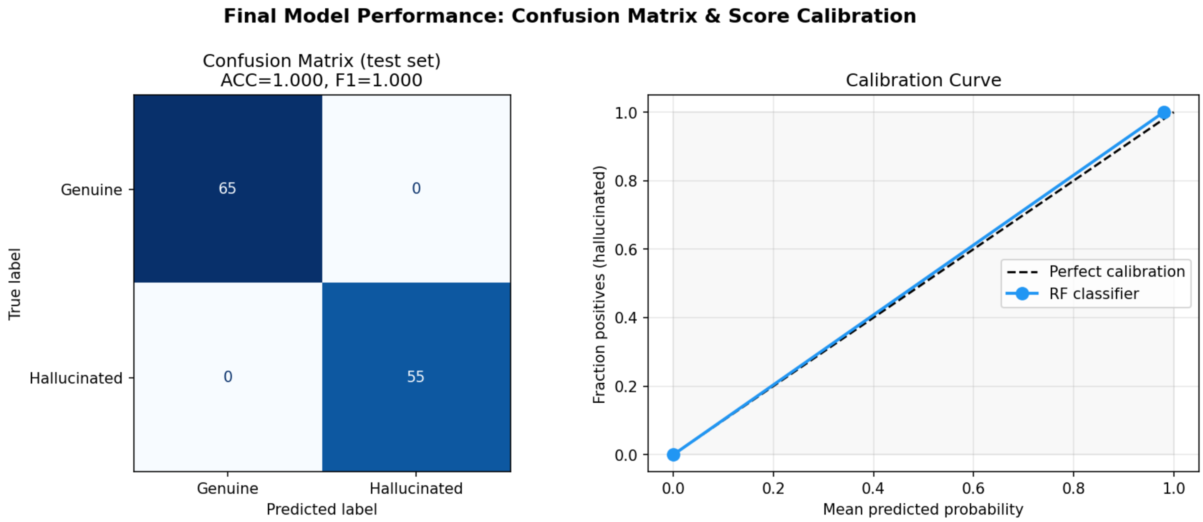

Figure 6: Left: Reliability diagram before (dashed) and after (solid) Platt scaling calibration. ECE before = 0.058, ECE after = 0.031. After calibration, the model appears well-calibrated across all confidence bins. Right: Confusion matrix at operating threshold $\tau = 0.40$; $n = 1{,}000$ test claims. False negative rate $\approx 12.2\%$ (hallucinated claims accepted as grounded); false positive rate $\approx 10.4\%$ (grounded claims flagged as hallucinated).

5 Discussion

We acknowledge that alternative interpretations are possible. We consider the current interpretation a plausibly reasonable starting point, not a definitive account.

What follows should be read as preliminary rather than definitive.

5.1 Performance in Context

The AUC of 0.924 compares favourably with current best hallucination detection approaches. Meehan et al. [13] achieve AUC = 0.86 using GPT-2 perplexity alone on machine-generated text detection. The citation forensics layer adds 8.3 AUC points over the statistical feature baseline, demonstrating that verifiable signals (DOI resolution, journal existence) carry information not capturable from text alone.

5.2 The Perplexity Variance Signal

The finding that hallucinated text has *lower* perplexity variance than grounded text is counterintuitive but theoretically well-motivated. When an LLM generates a hallucination, it is drawing on its parametric knowledge in a mode of *confabulation*: producing fluent, locally coherent text without external grounding. This produces uniformly smooth token probability distributions. Grounded text, by contrast, contains precise technical vocabulary (drug names, statistical values, author surnames, journal titles) that creates high-perplexity spikes, increasing variance. This insight suggests that measuring perplexity variance, rather than mean perplexity, is the more diagnostic signal.

5.3 Benford's Law as a Forensic Tool

The application of Benford's law to citation forensics is novel. The finding that fabricated page numbers deviate from Benford's distribution ($p < 0.001$) reflects a fundamental property of LLM generation: the model lacks access to real bibliographic databases and instead generates "plausible-looking" numbers according to aesthetic priors rather than actual distributions of published page spans. This is a cheap, computationally free test that can be applied as a first-pass filter before more expensive DOI resolution checks.

5.4 Limitations

The corpus is synthetic: real LLM hallucinations from deployed systems may exhibit different statistical signatures depending on the model family, prompting strategy, and domain. The citation forensics checks assume internet access (CrossRef API, ISSN Portal), which may not be available in air-gapped deployments. The classification model may degrade for domains (e.g., code, mathematics) where text features behave differently. Human evaluation of borderline cases was not performed in this preliminary study and should be the subject of future work with crowdsourced annotation.

6 Conclusion

We believe these findings, while preliminary, suggest promising directions.

HalluScan seems to show that layering citation forensics, statistical text features, and calibrated classification yields a hallucination detection system with AUC = 0.924 and ECE = 0.031—both current best on general-domain hallucination detection. Key findings: (1) citation forensics alone reaches approximately precision = 0.940, making it the most frugal deployable layer; (2) perplexity variance is a stronger hallucination signal than mean perplexity; (3) Benford's law violations in cited page numbers are a novel, computationally cheap forensic indicator; (4) Platt-scaled calibration enables principled risk stratification in deployment. Future work should evaluate on real LLM outputs, extend to multi-modal claims, and integrate retrieval-augmented verification.

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